



7831 - Development of a Time-Variable Bad Pixel Mask for MIRI LRS from Blank Sky and Failed Observations

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The bad pixel mask for MIRI LRS is currently constant across all observations. However, it has become apparent that defective pixels have accumulated over time and are not being properly handled by the data reduction pipeline. These defective pixels produce apparent features in extracted spectra that are comparable in scale to known astrophysical features. Consequently, these features may result in spurious results when extracted spectra are analysed. We aim to produce a time-variable bad pixel mask that correctly flags these accumulating defective pixels for handling by the JWST pipeline.

The data we aim to use in our development are blank deep-sky observations with MIRI LRS. We are aware of multiple programs containing observations that, for various reasons, do not include any targets. These observations produce deep 2D detector readout images of blank regions of sky, within which defective pixels are readily visible. Through simple statistical analysis of detector pixels in the context of their environments, significant outliers corresponding to defective pixels can be easily flagged. With the full suite of blank-sky observations over an extended period of time, we can produce a time-dependent bad pixel mask that will significantly improve the handling of defective pixels in MIRI LRS. This could be extended to MIRI MRS and imaging modes, and potentially to NIRSpec modes as well.

OBSERVING DESCRIPTION